

UMAR HAMZAH

hmzhumar@gmail.com | 089518065450 | South Jakarta, DKI Jakarta | linkedin.com/in/umar-hamzah-si

PROFILE SUMMARY

First-batch Information Systems student at Universitas Cakrawala with a technical foundation from SMK Telkom Jakarta in Software Engineering. Experienced in IoT development and robotics education through two internships. Participated in Gemastik XVIII 2025, a national ICT competition by Indonesia's Ministry of Education, developing a smart pilgrim coordination platform using mesh networking and machine learning. Skilled in web development, embedded systems, UI/UX design, and AI-based anomaly detection. A collaborative team player with strong problem-solving skills and a passion for continuous learning.

EDUCATION

B.Sc. Information Systems — Universitas Cakrawala (Founding Batch) | 2024 – Present

- Selected as participant at Gemastik XVIII 2025 (Smart Devices category) — national ICT competition by Ministry of Education x Universitas Telkom
- Actively building cross-subject projects spanning web, IoT, mobile, and AI domains

Vocational High School — Software Engineering — SMK Telkom Jakarta | July 2021 – June 2024

INTERNSHIP EXPERIENCE

IoT Developer Intern | PT Global Inovasi Terdepan (RightJet) 2025

- Developed an internet-free offline communication system using ESP32 as an alternative device for Umrah and Hajj pilgrims.
- Designed an ESP32-based primary channel connecting directly to pilgrims' smartphones, replacing conventional communication tools.
- Applied embedded systems and wireless communication concepts in a hardware-constrained environment.

Assistant Teacher — Robotics Education | PT Sari Teknologi July 2023 – December 2023

- Assisted robotics learning sessions across 4 schools, from elementary to junior high school level.
- Designed and prepared teaching materials tailored to each class's ability level based on prior assessment.
- Developed IoT competency through a group project focused on IoT-based system development.

PROJECT EXPERIENCE

Sistem Koordinasi Jamaah Adaptif — Gemastik XVIII 2025 (Smart Devices Category) October 2025
Universitas Cakrawala | Stack: ESP32-WROOM-32, GPS NEO-8M, ESP-NOW Mesh, DBSCAN, One-Class SVM, MapLibre, Arduino IDE

- Designed a hybrid dual-mode IoT platform for Umrah/Hajj pilgrim coordination operating offline via ESP-NOW mesh network and online via cloud sync.
- Implemented two-layer anomaly detection (DBSCAN + One-Class SVM) achieving 94% accuracy and only 6% false alarm rate.
- Built offline real-time map, SOS emergency button, Push-to-Talk voice, and digital Tawaf & Sai counter. Network initialized in 25 seconds.

TrakPoint — Smart Fleet Monitoring System Semester 2, 2024
Web Client Development | UX Design | Database | Stack: PHP, HTML, CSS, MySQL, Node.js, WebSocket, Mosquitto (MQTT), ESP-32, GPS NEO-6M, Figma

- Built a web-based truck fleet monitoring system with real-time GPS tracking, automatic overspeed alarm, idle detection, and trip history logging.
- Integrated IoT hardware (ESP-32, GPS NEO-6M, SIM800L) for location detection and data transmission to the web dashboard.
- Designed UI/UX in Figma and built a MySQL database schema with multi-role user system (Admin / Operator / Driver).

TransportKu — Smart City Transportation Assistant Semester 3, 2025
Information Systems Concept | Stack: Figma, Use Case Diagram, BPMN, Activity Diagram

- Conceptualized an integrated transportation platform combining train/bus ticketing and ride-hailing in a single application.
- Produced complete system documentation (Use Case, BPMN, Activity Diagram) and designed 30+ UI/UX screens in Figma.

TECHNICAL SKILLS

Programming Languages: HTML, CSS, PHP, JavaScript, C++, SQL

Tools & Software: Figma, Arduino IDE, VS Code, Node.js, MapLibre, Mosquitto (MQTT), WebSocket, Tableau, Canva, Microsoft Office

Hardware / IoT: ESP-32, ESP-NOW Mesh, GPS NEO-6M, GPS NEO-8M, SIM800L, RTC Module

AI / Algorithms: DBSCAN, One-Class SVM, Anomaly Detection

Soft Skills: Teamwork, Communication, Problem Solving, Time Management, Adaptability, Fast Learner

CERTIFICATIONS

Certificate of Appreciation — IoT Developer Intern | PT Global Inovasi Terdepan (RightJet) | November 2025

Certificate of Participation — GEMASTIK XVIII (Smart Devices) | Ministry of Education x Universitas Telkom | October 2025

Certificate of Competence — Junior Coder Occupation (Software & Game Development) | BNSP (Indonesian Professional Certification Authority) | May 2024

Vocational Competency Certificate — Web-Based School Complaint App (PHP, MySQL, Bootstrap) | SMK Telkom Sandhy Putra Jakarta | Score: 89/100 (Highly Competent) | June 2024

Vocational Competency Certificate — RFID Implementation on Delivery Robot | SMK Telkom Sandhy Putra Jakarta | Score: 90/100 (Competent) | January 2024

LANGUAGES

Indonesian — Native **English** — Intermediate | TOEIC Score: 415 (Listening: 235, Reading: 180) | ETS, September 2023